

GradeWise Service

Automated Exam Evaluation using AI

✓ Question Paper > ✓ Model Answer > ✓ Student Sheet > 4 Results

Start New Evaluation

Evaluation Result

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Student ID: MH003

31.0 / 40.0

Scoreboard

Question	Score	Max	%		Status
Section A					
Q1	1	1	100%	1/1	Scored
Q2	1	1	100%	1/1	Scored
Q3	1	1	100%	1/1	Scored
Q4	1	1	100%	1/1	Scored
Q5	1	1	100%	1/1	Scored
Q6	1	1	100%	1/1	Scored
Q7	1	1	100%	1/1	Scored

Question	Score	Max	%		Status
Q8	0	1	0%	0/1	Not Scored
Section B					
Q9	2	2	100%	2/2	Scored
Q10	2	2	100%	2/2	Scored
Q11	2	2	100%	2/2	Scored
Q12	2	2	100%	2/2	Scored
Q13	2	2	100%	2/2	Scored
Q14i	0	1	0%	0/1	Not Scored
Q14ii	0	1	0%	0/1	Not Scored
Section C					
Q15	1.5	3	50%	1.5/3	Scored
Q16i	0	1.5	0%	0/1.5	Not Scored
Q16ii	0	1.5	0%	0/1.5	Not Scored
Q17	3	3	100%	3/3	Scored
Q18	1.5	3	50%	1.5/3	Scored
Section D					

Question	Score	Max	%	Status
Q19	8	8	100%	8/8 Scored
Total	31	40	78%	— —

Question-wise Details

Question 1

1 / 1

The student's answer matches the model answer exactly, selecting option (b) 8, which is the correct number of subsets of the set $A = \{1, 2, 3\}$.

Question 2

1 / 1

The student's answer matches the model answer exactly in content and meaning, correctly identifying the universal set as the set containing all elements under consideration. The slight difference in punctuation (a period at the end) does not affect correctness.

ⓘ Suggestions

- Ensure to follow the exact format requested, but minor punctuation differences are acceptable.
- Continue to provide clear and precise answers as demonstrated.

Question 3

1 / 1

The student's answer matches the model answer exactly. Since $\sin A = 3/5$ and A is acute, $\cos A = 4/5$ is correct, corresponding to option (a). Full marks are awarded.

Question 4

1 / 1

The student's answer '(c) Many one relation.' matches the model answer '(c) Many-one relation' exactly in meaning. The slight difference in punctuation (hyphen vs

space) does not affect correctness. Therefore, full marks are awarded.

🚩 **Suggestions**

- Use a hyphen in 'Many-one' to match the standard terminology.
- Avoid adding a period at the end of the answer in MCQ selections.

Question 5

1 / 1

The student's answer matches the model answer exactly, both indicating that $\sin 30^\circ = 1/2$. Therefore, full marks are awarded.

Question 6

1 / 1

The student's answer matches the model answer exactly, both selecting option (c) 60 as the number of ways to arrange 3 books out of 5. This is the correct answer, so full marks are awarded.

🚩 **Suggestions**

- No improvements needed as the answer is correct and complete.

Question 7

1 / 1

The student's answer matches the model answer exactly, correctly identifying that if A and B are disjoint sets, then their intersection is the empty set ($A \cap B = \emptyset$).

🚩 **Suggestions**

- No improvements needed as the answer is correct and complete.

Question 8

0 / 1

The correct answer is (c) Transitive. The student's answer (a) Reflexive is incorrect because the 'less than' relation on natural numbers is not reflexive; no number is less than itself. Therefore, the student receives zero marks.

🚩 Suggestions

- Review the definitions of reflexive, symmetric, and transitive relations.
- Understand that 'less than' is a strict order relation which is transitive but not reflexive or symmetric.
- Practice identifying properties of relations on sets to improve accuracy in such questions.

Question 9

2 / 2

The student's answer correctly defines a set as a well-defined collection of objects and provides a valid example $A = \{1, 2, 3\}$. The slight wording difference ('very well-defined' instead of 'well-defined') does not affect correctness or completeness.

🚩 Suggestions

- Use standard phrasing such as 'a well-defined collection of objects' to maintain clarity.
- Add a period after 'objects' to improve punctuation.
- Capitalize 'Example' to match standard formatting.

Question 10

2 / 2

The student's answer correctly identifies the domain as all real numbers ($\mathbb{R}$) and the range as all real numbers greater than or equal to zero, expressed as $[0, \infty)$. This matches the model answer's domain (all real numbers) and range ($x \geq 0$). The notation used by the student is clear and mathematically correct.

🚩 Suggestions

- Use set-builder notation consistently if required by the question, e.g., $\{x : x \geq 0\}$.
- Clarify that the domain is all real numbers by explicitly stating 'all real numbers' or using the symbol $\mathbb{R}$.

Question 11

2 / 2

The student correctly evaluated $\sin 45^\circ \cos 45^\circ$ by expressing both sine and cosine of 45 degrees as $1/\sqrt{2}$ and multiplying them to get $1/2$. The answer matches the

model answer exactly, including the correct final numerical value and the complete steps.

🚩 **Suggestions**

- Ensure consistent use of lowercase for trigonometric functions (sin, cos) as per standard mathematical notation.
- No further improvements needed as the answer is complete and accurate.

Question 12

2 / 2

The student correctly calculated the value as 20 with the correct multiplication steps shown. Although the notation ' 5P_2 ' is incorrect for ' 5P_4 ', the numerical value and calculation correspond to the model answer's value of 20. Therefore, full marks are awarded.

🚩 **Suggestions**

- Use the correct permutation notation matching the question, i.e., ' 5P_4 ' instead of ' 5P_2 '.
- Ensure that the number of permutations calculated corresponds exactly to the problem statement.

Question 13

2 / 2

The student correctly identified that the number of elements in the power set of a set with 4 elements is $2^4 = 16$. The answer is both correct and complete.

🚩 **Suggestions**

- The student could improve clarity by explicitly stating that the power set contains all subsets of the original set.
- Using consistent notation for exponents (e.g., '^' or superscript) can enhance readability.

Question 14i

0 / 1

No student answer available.

 Suggestions

- Provide an answer to receive credit.

Question 14ii

0 / 1

No student answer available.

 Suggestions

- Provide an answer to receive credit.

Question 15

1.5 / 3

The student correctly identified the left-hand side expression and stated that by taking the LCM and simplifying, it equals the right-hand side ($2 \operatorname{cosec}\theta$). However, the student did not show any intermediate algebraic steps or the detailed simplification process that leads to the final result. Since the final conclusion is correct but lacks the necessary working, half marks are awarded.

 Suggestions

- Show all intermediate steps when simplifying expressions, such as combining terms over a common denominator and expanding squares.
- Explicitly write out the algebraic manipulations to demonstrate how the left-hand side simplifies to the right-hand side.
- Use consistent notation (e.g., lowercase 'sin' and 'cos') and clearly indicate each step to improve clarity and completeness.

Question 16i

0 / 1.5

No student answer available.

 Suggestions

- Provide an answer to receive credit.

Question 16ii

0 / 1.5

No student answer available.

 Suggestions

- Provide an answer to receive credit.

Question 17

3 / 3

The student correctly identified the number of 3-digit numbers as $5P3$ and showed the complete calculation $5 \times 4 \times 3 = 60$. The reasoning, though brief, is sufficient and matches the model answer in correctness, completeness, and accuracy.

 Suggestions

- Provide a slightly more detailed explanation of why permutations are used (e.g., because digits cannot be repeated and order matters).
- Clarify that the digits are chosen without repetition to reinforce understanding.

Question 18

1.5 / 3

The student correctly identified that the relation is reflexive, symmetric, and transitive, which matches the model answer's conclusion. However, the student did not provide any reasoning or explanation to support these claims. Since the final conclusion is correct but lacks any working or justification, half marks are awarded.

 Suggestions

- Provide reasoning or examples to show why the relation is reflexive, symmetric, and transitive.
- Explain that $(a,a) \in R$ for all elements a in A to demonstrate reflexivity.
- Show that for every (x,y) in R , (y,x) is also in R to prove symmetry.
- Demonstrate that whenever (x,y) and (y,z) are in R , (x,z) is also in R to establish transitivity.

Question 19

8 / 8

The student correctly applied the fundamental trigonometric identity $1 + \tan^2\theta = \sec^2\theta$ to transform the left-hand side expression into the right-hand side expression. The steps are clear and logically presented, leading directly to the conclusion that LHS = RHS. Although the answer is brief, it includes the key identity and the necessary reasoning to prove the given identity fully.

🚩 **Suggestions**

- Provide a bit more detail by explicitly stating the substitution step to enhance clarity.
- Include a concluding statement such as 'Hence proved' to clearly indicate the completion of the proof.